

Bias and Fairness Report

This report outlines the **methodological limitations, proxy assumptions, and error distributions** identified within our analytical framework. Acknowledging and addressing these systemic gaps ensures structural transparency and helps mitigate risks before production deployment.

1. Sampling and Coverage Limitations

Our dataset exhibits several **structural sampling biases** that prevent it from perfectly representing the broader target population.

- **Underrepresented Demographics (Coverage Gap):** The data collection pipeline heavily favors urban centres and tech-literate populations due to a reliance on digital convenience sampling. Rural populations and individuals over the age of 60 are statistically underrepresented.
- **Attrition and Non-Response Bias:** In our longitudinal datasets, we observed non-random attrition. Users from lower socio-economic backgrounds dropped out at a higher rate, skewing the final analytical sample towards higher-income brackets.
- **Scale Fallacy:** Increasing the volume of data did not correct these skewness patterns; a larger sample size merely generated a more mathematically precise estimate of a fundamentally biased baseline.

2. Uneven Error Distribution

An evaluation of top-line metrics indicated stable global accuracy; however, a deep-dive **subgroup performance audit** revealed that errors are distributed unevenly across conditions and timelines.

Conditional Error Disparities

The model exhibits a disproportionately high **False Negative Rate (FNR)** for minority groups and regional variations. Because the optimization function prioritizes global accuracy, the model defaults to majority-population patterns. Consequently, anomalous but valid behaviours among minority subsets are flagged as errors or outliers far more frequently than majority behaviours.

Temporal Drift and Seasonal Variance

Errors are not static across time. We observed significant spikes in model inaccuracies during specific periods:

- **Seasonal Volatility:** During holidays and major promotional cycles, user behaviour diverges from the baseline training distribution, causing a temporary spike in **False Positive flags**.
- **Macroeconomic Shifts:** The model lacks real-time elasticity. Sudden external market changes degrade the reliability of older historical data, forcing a rapid increase in systemic error rates if the model is not continuously re-calibrated.

Mobility Governance and Sustainability in AI (Mobility AI) Systems Both of these elements are extremely important for the responsible use of the model.

Both these elements are briefly discussed below:

1. Governance & Oversight required before use of the model

The following safety and monitoring measures must be in place before an AI model can be used for actual traffic or mobility control:

- **Rigid Testing & Validation:** The model needs to be thoroughly tested to ensure it works accurately in different weather conditions, during rush hours, and in unexpected situations (edge cases).
- **Safety and Human-in-the-Loop:** In critical or emergency situations, there must be a system of human oversight to make final decisions, so that accidents due to AI errors can be avoided.
- **Transparency and Explainable AI:** The explanation of why the model made a decision (e.g. changing direction or braking) should be easy for the administrator to understand.
- **Data Privacy & Cybersecurity:** Passengers' personal data and travel information should be completely secure. Also, the model should be safe from cyber attacks (Hacking).
- **Bias & Fairness Evaluation:** It is necessary to ensure that the AI model does not discriminate against people from certain regions or specific classes.

2. Environmental & Resource Trade-offs

Building and running dynamic AI models has the following impacts on the environment and resources, which need to be balanced:

- **Compute Power vs. Accuracy:**
 - **Problem:** To achieve greater accuracy, complex models like 'Deep Learning' have to be used. This requires a large amount of data centers and electricity, which increases carbon emissions.

- **Alternative:** It is often beneficial for the environment to choose simpler models (e.g. Edge AI or Light Models) that provide 95% accuracy but run on very low energy instead of 99% accuracy.
- **Hardware & Resource Limitations (Hardware & E-Waste):**
 - **Problem:** High-end models require expensive GPUs and constant new hardware to run, which increases e-waste.
 - **Alternative:** Resources can be saved by using algorithms (e.g. Model Quantization) that can run on older or limited hardware.
- **Real-time Processing Trade-off:**
 - For real-time traffic control, processing data locally (on edge devices) instead of sending it to the cloud saves both network bandwidth and energy.